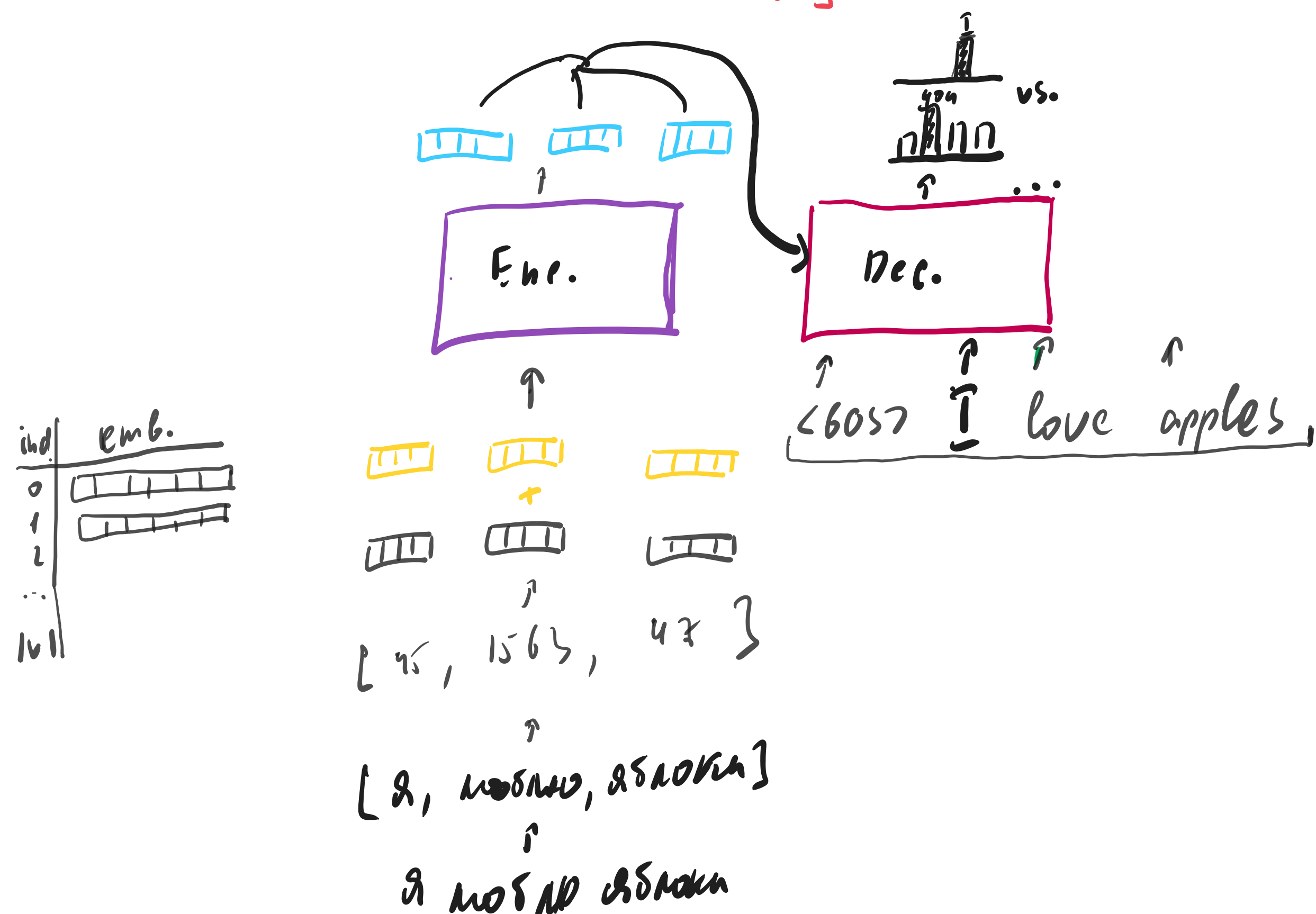


- Recap
- How to generate

target: I love apples less



## Encoder

- Self-Attn
- FFN

## Decoder

- Masked Self-Attn
- Encoder-Decode Attn
- FFN

Encoder-decoder attn.

$$z = \text{softmax} \left( \frac{Q K^T + \text{Mask}}{\sqrt{d}} \right) V$$

$$\text{Mask} = \begin{bmatrix} 0 & -\text{inf} & -\text{inf} & \dots \\ 0 & 0 & -\text{inf} & \dots \\ 0 & 0 & 0 & \dots \\ \dots & \dots & \dots & \dots \end{bmatrix}$$